

CAMBRIDGE INTERNATIONAL EXAMINATIONS - BIOLOGY 9700/12

Paper 1 Multiple Choice - May/June 2023

- 1 Which organelle is responsible for ATP synthesis during aerobic respiration?
A Nucleus B Mitochondrion C Ribosome D Golgi apparatus [1]
- 2 What is the primary mechanism of ultrafiltration in the glomerulus of human kidneys?
A Active transport across podocyte membranes B Hydrostatic pressure differential
C Osmotic gradient created by loop of Henle D Facilitated diffusion [1]
- 3 Which protein structure level is stabilized by disulfide bridges between cysteine residues?
A Primary structure B Secondary alpha-helix C Tertiary structure D Peptide backbone [1]
- 4 What change occurs in the sinoatrial node when parasympathetic stimulation increases?
A Hyperpolarization via increased K⁺ permeability B Increased rate of Na⁺ influx
C Depolarization of cardiac muscle fibers D Accelerated AV node conduction [1]
- 5 Which enzyme catalyzes the conversion of carbon dioxide and water to carbonic acid in RBCs?
A Carbonic anhydrase B Hemoglobin reductase C Carboxypeptidase D Catalase [1]